

A Simulated Hybrid PQC–QKD Edge Computing Framework for Secure and Low-Latency Internet of Medical Things (IoMT)

1 Team Details

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2 Introduction and Background

The rapid proliferation of the Internet of Medical Things (IoMT) has revolutionized real-time patient monitoring, enabling wearable sensors to transmit critical physiological data (e.g., electrocardiograms, glucose levels) to healthcare datacenters. However, these networks rely on classical cryptographic algorithms (RSA, ECC) whose security relies on the hardness of prime factorization and discrete logarithms. With the imminent arrival of cryptographically relevant quantum computers running Shor’s algorithm, current IoMT networks face critical vulnerabilities.

To achieve quantum resistance, the research community is advancing two distinct paradigms: Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC). QKD offers unconditional, physics-based security but requires bulky hardware and fiber-optic links, making it physically impossible to deploy on wearable medical sensors. Conversely, PQC offers software-based mathematical resistance (e.g., lattice-based cryptography) but imposes extreme computational overheads, risking battery depletion and fatal transmission delays for time-sensitive clinical alerts. This project introduces a hybrid, three-tier architecture bridging these two paradigms.

3 Literature Gap & Problem Statement

3.1 The Literature Gap (Data-Backed Analysis)

Based on an extensive review of IEEE Transactions from 2025–2026, several critical gaps remain in securing edge-enabled IoMT networks:

- **The PQC Computational Bottleneck:** Deploying pure Post-Quantum Cryptography (e.g., ML-KEM/Kyber) on edge IoMT devices introduces severe computational bottlenecks. Lattice-based discrete Gaussian sampling imposes unsustainable

overheads on low-power devices. Unoptimized baselines (e.g., Cao et al., Saeed et al.) require up to **570.3 ms for encryption** and **186.7 ms for decryption**. This massive delay entirely violates the strict < 250 ms real-time latency threshold required for critical cardiac telemetry.

- **The PQC Communication Overhead:** PQC algorithms suffer from massive key storage demands. Transporting large public/private keys across limited wireless channels results in excessive network congestion. Empirical data shows that pure PQC models consume approximately **2.4 mJ** per transmission, rapidly draining wearable batteries.
- **Lack of Tiered Hybrid Frameworks:** Current state-of-the-art literature explores PQC and QKD in isolation. There is a lack of simulation-backed frameworks that strategically distribute these cryptographic workloads across varying hardware tiers to balance security, latency, and energy constraints.

3.2 Proposed Problem Statement

*“To design, simulate, and benchmark a tiered **Hybrid PQC–QKD Edge Computing Framework for IoMT** that resolves the quantum computational bottleneck on edge devices. By restricting lightweight Post-Quantum Cryptography (ML-KEM) to short-range wearable-to-edge hops and offloading long-range backbone transmission to a simulated Quantum Key Distribution (BB84) channel, this project aims to prove empirically that the architecture bypasses the severe delays of pure PQC, staying within real-time clinical safety thresholds (< 250 ms) while ensuring absolute quantum-resistance and preserving wearable energy constraints.”*

4 Proposed Hybrid Architecture and Workflow

The project proposes a novel three-tier deterministic simulation pipeline:

1. **Tier 1: IoMT Wearable (PQC Domain):** The patient’s wearable device generates synthetic vitals. Due to battery constraints, it utilizes a computationally optimized Post-Quantum algorithm (NIST FIPS 203 ML-KEM) to encapsulate a symmetric key. The data is transmitted over a simulated short-range wireless channel to the room’s edge router.
2. **Tier 2: Edge Node (The Hybrid Gateway):** The hospital ward’s Edge Node acts as a cryptographic bridge. It uses its PQC private key to decapsulate the data, aggregates payloads from multiple simulated patients, and acts as a QKD transmitter for the next hop.
3. **Tier 3: Healthcare Datacenter (QKD Domain):** The Edge Node and Datacenter perform a simulated BB84 QKD protocol over an optical fiber link. If the

Quantum Bit Error Rate (QBER) remains below the safety threshold, the perfectly secure key is used to encrypt the aggregated batch via AES-256 for the final long-haul transmission.

5 Mathematical Modeling and Formulations

To ensure rigorous performance evaluation, the simulation is anchored by the following primary mathematical models extracted from current literature:

5.1 Lattice-Based Encryption Overhead (PQC)

The computational cost at the edge is modeled using the Learning With Errors (LWE) lattice formulation:

$$\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q} \quad (1)$$

where $\mathbf{A}$ is a uniformly random matrix, $\mathbf{s}$ is the secret vector, $\mathbf{e}$ is the Gaussian noise vector, and q is the prime modulus. The time complexity $\mathcal{O}(n \log n)$ of the polynomial multiplication will be benchmarked via CPU cycle counting.

5.2 Total System Latency Model

The total end-to-end latency (T_{sys}) constraint for the clinical telemetry is formulated as:

$$T_{sys} = T_{PQC}^{enc} + T_{link}^{short} + T_{Edge}^{process} + T_{QKD}^{gen} + T_{link}^{long} \leq 250 \text{ ms} \quad (2)$$

5.3 Quantum Bit Error Rate (QBER) Constraint

For the Tier 2 to Tier 3 backbone, the protocol security against eavesdropping is governed by the QBER:

$$QBER = \frac{N_{error}}{N_{total}} = \frac{P_{dark} + P_{opt}}{P_{dark} + P_{opt} + P_{signal}} \quad (3)$$

where P_{dark} is detector dark count probability, and P_{signal} is the received photon probability governed by fiber attenuation (0.2 dB/km). If $QBER > 11\%$, the transmission is aborted.

6 Simulation Setup and Parameters

The project will be virtually implemented entirely in software without requiring physical quantum hardware.

- **Simulation Stack:** Python 3.10+, `liboqs` (Open Quantum Safe wrapper) for ML-KEM integration, IBM `Qiskit` for quantum photon polarization logic, and `Mininet` for network topology emulation.
- **Network Parameters:**

- *Edge Link*: Bandwidth 35 MHz, 5–10 ms base delay, packet sizes ranging 512 Bytes to 2 KB.
- *Backbone Link*: Modeled with 50 km physical distance, 0.2 dB/km attenuation, 128 kbps quantum emission rate.
- **Evaluation Metrics**: The final phase will produce comparative graphs evaluating the Hybrid Architecture against Legacy RSA and Pure End-to-End PQC frameworks based on **End-to-End Latency (ms)**, **Edge Node CPU Cycle Count**, and **Estimated Energy Consumption (mJ)**.

7 Base Papers (IEEE Transactions, 2025–2026)

- [1] **Variational Quantum Reinforcement Learning for Joint Resource Allocation of Blockchain-Based Vehicular Edge Computing and Quantum Internet** (*IEEE Trans. on Vehicular Technology*, 2025) — Provides the Markov Decision Process (MDP) modeling for latency optimization, forming the basis for our T_{sys} evaluation metrics.
- [2] **An Efficient and Robust Resource Allocation Method for Quantum Key Distribution Networks** (*IEEE Trans. on Network and Service Management*, 2025) — Provides the Integer Linear Programming (ILP) mathematical model for quantum key capacity constraints and link attenuation modeling.
- [3] **Post-Quantum Cryptography (PQC) for IoT-Consumer Electronics Devices Integrated With Deep Learning** (*IEEE Trans. on Consumer Electronics*, 2025) — Establishes the empirical baseline of 25.3 ms latency and 2.4 mJ energy consumption for PQC edge operations, acting as the primary benchmark we aim to improve upon.
- [4] **Post-Quantum Identity-Based Linkable Ring Signcryption for Edge IoT Devices** (*IEEE Trans. on Consumer Electronics*, 2026) — Supplies the complex lattice-generation models ($\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q}$) and discrete Gaussian sampling formulas used for quantifying wearable processing overhead.
- [5] **Post-Quantum Traceable Anonymous Cross-Domain Authentication for Blockchain-Based IoT** (*IEEE Trans. on Consumer Electronics*, 2026) — Supplies the architectural baseline data for evaluating cross-domain cryptographic bridging and aggregation computation at the edge node (0.57 ms processing baseline).